



Determining the absolute abundance of dinoflagellate cysts in recent marine sediments III: Identifying the source of *Lycopodium* loss during palynological processing and further testing of the *Lycopodium* marker-grain method

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ABSTRACT

Lycopodium clavatum marker-grain tablets are often added before palynological processing to calculate the absolute abundance of organic-walled microfossils in sediments. In this study we investigate the reproducibility of dinoflagellate cyst concentrations by performing nine different tests: varying dinoflagellate cyst/*Lycopodium* spore ratios, sample weights, centrifuge speeds and times, sonication times, and mesh sizes; manually powdering the sediment using a mortar and pestle prior to processing, using a water jug instead of squirt bottles during sieving, decanting over a 6 µm sieve and adding *Lycopodium* tablets at the end. Previous studies have reported loss of *Lycopodium* spores during processing. In this study we investigate sources of this loss by microscopically analyzing residue on stir sticks and sieves, and what is lost through the sieves and decanting. We document an average loss of 8.2 % of *Lycopodium* spores during processing, with the greatest loss occurring through the fine sieve and resulting in only a 4.8% overestimation of the cyst concentration. Results using powdered sediment and sonicating for more than 5 min are less reproducible. Large cyst/*Lycopodium* spore ratios were found to have higher cyst concentrations and standard deviations, thus we recommend ratios of less than five, and ideally ratios between 0.5 and 2 to facilitate counting.

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1. Introduction

Determining the precise absolute abundance of dinoflagellate cysts in sediments is important for both recent environmental studies (e.g., Dale, 1996; Bringué et al., 2013; Limoges et al., 2013; Heikkilä et al., 2014) and paleoceanographic studies (e.g., Pospelova et al., 2006; Holzwarth et al., 2010; Marret et al., 2013; Price et al., 2013). To concentrate dinoflagellate cysts, analysts use various palynological treatment methods, which often involve the use of hydrochloric (HCl) acid to dissolve carbonates, hydrofluoric (HF) acid to dissolve silicates, and sieving to eliminate the coarse and fine fractions (e.g., Mertens et al., 2009). Oxidizing reagents, such as nitric acid and hydrogen peroxide, are known to damage or destroy dinoflagellate cysts (e.g., Reid, 1977; Marret, 1993; Hopkins and McCarthy, 2002; Zonneveld et al., 2007) and should be avoided.

Concentrations and fluxes of dinoflagellate cysts are calculated by adding a known number of exotic markers or “spike” to each sediment

sample (e.g., Stockmarr, 1971). It allows the analyst to count only a proportion of the sample, and by using the number of cysts counted as well as the ratio of marker grains counted to marker grains added, absolute abundances can be calculated (see Section 2.2). The most commonly used marker are spores of the plant *Lycopodium clavatum* (wolf's-foot clubmoss, stag's horn clubmoss, or ground pine). The spore of *L. clavatum* is trilete, and ~20–40 µm in diameter (Lignum et al., 2008). A *Lycopodium* tablet is produced in such a way that it contains an estimated number of spores, commonly in the tens of thousands depending on the batch number, and a standard deviation. The tablet matrix is mainly sodium carbonate or sodium bicarbonate together with polyethyleneglycol. The spores have been acetolysed, giving them an orange color that allows them to be distinguished from natural *Lycopodium* spores. *Lycopodium* tablets are produced by the Department of Quaternary Geology, University of Lund, Lund, Sweden.

While the number of cysts counted is sometimes provided in Supplementary material (e.g., Chen et al., 2011), the number of *Lycopodium* spores counted is only occasionally reported (e.g., Head, 1993; Pospelova et al., 2004; Limoges et al., 2013). The ratio of cysts counted to *Lycopodium* spores counted is also commonly ignored. Maher (1981) suggests that

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the ratio between the number of pollen counted (or in our case cysts) to the number of *Lycopodium* spores counted should be about two in order to maximize information about the fossil assemblage, maintain precise concentration estimates, and reduce counting time. However, this ratio has not been verified experimentally. In this study, we aim to optimize the use of this cyst/*Lycopodium* spore ratio and to determine how cyst concentrations and their standard deviations vary if this ratio is smaller or larger than two.

Recent inter- and intra-laboratory studies by Mertens et al. (2009, 2012) demonstrate how different palynological processing techniques can result in different calculated dinoflagellate cyst concentrations. As Mertens et al. (2009) show, there is no “standard” palynological treatment method that is universally practiced. Instead, the volume and temperature of acids (HCl and HF) vary, as well as the type of sieve (metal, nylon, polyester), the size of sieve (6–20 µm), centrifuge time (5–600 s), centrifuge speed (1900–3500 rpm), the practice of pre-sieving, and the length of sonication time (0–1800 s) (Mertens et al., 2009, Supplementary material). Mertens et al. (2012) analyzed the effect of using acids at different temperatures, of using metal versus nylon sieves, and the use of pre-sieving. They found that metal sieves yielded more consistent results and that using HF at 40 °C or 60 °C for 30 + min significantly affects the reproducibility of cyst concentrations. They also found that calculated cyst concentrations were significantly higher in samples where the *Lycopodium* tablets were added prior to processing than in samples where the tablets were added at the end (Mertens et al., 2012). As no *Lycopodium* spores can be lost when the tablets are added at the end, this suggests some are lost during processing.

In this paper, we document when *Lycopodium* spores and dinoflagellate cysts are lost during processing and study the effects of different palynological processing techniques on the reproducibility of cyst concentrations in surface sediments collected from Saanich Inlet (British Columbia, Canada). The effects tested here are (1) varying the cyst/*Lycopodium* spore ratio; (2) varying the sediment weight and keeping the cyst/*Lycopodium* spore ratio constant; (3) powdering the sediment prior to analysis; (4) using a water pitcher instead of squirt bottles during sieving; (5) using 10 µm instead of 15 µm sieves; (6) sonicating for different lengths of time; (7) varying centrifuge times and speeds, not using a centrifuge but settling for 7 to 24 h; (8) decanting over a 6 µm sieve; and (9) adding the *Lycopodium* tablet at the end of processing. We make recommendations to further optimize the *Lycopodium* marker-grain method.

2. Materials and methods

2.1. Palynological processing

Surface sediments were obtained by V.P. from Patricia Bay in Saanich Inlet (British Columbia, Canada) using a Van Veen grab in summer 2012 (Lat. 48° 35.554'N, Long. 123° 30.034'W; water depth 226 m), close to the site PB3 studied by Mertens et al. (2012). The sediment was predominantly silt, with some fine sand and small pebbles up to ~3 mm in size. The sediment was stored in the dark at 4 °C prior to sampling.

Nine samples processed using the method outlined below are referred to in this paper as the reference set, and it is the method to which the other tests are compared. The samples were all processed by A.M.P. at the Paleoenvironmental/Marine Palynology Laboratory, University of Victoria, Canada. All samples are stored at the same location. To increase the statistical robustness, three samples were processed for each test. This allows us to calculate standard deviations and compare averages and ranges of values between tests. Unless otherwise noted the following methodology was used.

2.1.1. The reference set

The sediment was homogenized, freeze-dried, and weighed. As freeze-drying has been shown not to affect the absolute abundance of

dinoflagellate cysts (Mertens et al., 2012), we use this drying method to be consistent with previous experiments by Mertens et al. (2012). One gram of freeze-dried sediment was subsampled into 15 mL polypropylene centrifuge tubes. An average of 7–8 mL of room temperature 10% concentrated hydrochloric acid (HCl) was added for 15 min and stirred using a wooden stir-stick. The sediments contained a small amount of calcium carbonate, and all samples fizzed slightly in HCl when stirred. One *L. clavatum* tablet (batch no. 177745 = 18,584 spores/tablet, $\sigma = 1853$) was added to the HCl, and once the fizzing subsided, the solution was thoroughly mixed. Next samples were centrifuged using a centrifuge with a radius of 15 cm at 3000 rotations per minute (rpm) for 6 min, and the HCl was decanted.

The sediment was then rinsed twice with distilled water and sieved using 120 µm and 15 µm Nitex nylon stackable sieves. The 15 µm mesh (Nylon Nitex synthetic screen: NTX15, 10% open area, mesh opening size 15 µm, NITEX 03-15/5 made in Switzerland 2495720101) is made out of interwoven nylon strands. The width of each hole ranged from 12.1 to 17.8 µm with an average of 15.1 µm ($N = 22$) and had a hypotenuse of 15.7–23.6 µm, with an average of 20.2 µm ($N = 22$). A magnetic stir bar was placed on the 15 µm sieve and placed above a magnetic plate. The magnetic stir bar helps prevent the sediment from clogging up the sieve and speeds up the sieving process by approximately 2 to 4 h. Squirt bottles with distilled water were used to break up clay particles on the coarse sieve. The fraction collected on the 120 µm sieve was then poured into a glass beaker and sonicated using a Branson sonic bath for 20 s to disaggregate any remaining clumps and subsequently returned to the 120 µm sieve for further washing. When only the >120 µm material remained on the coarse sieve, it was then transferred to a vial. The remaining residue in the 15 µm sieve was rinsed with distilled water until the water draining from the sieve turned clear.

After sieving, the residue remaining on the 15 µm sieve was transferred back to the centrifuge tube, centrifuged for 6 min, and the water decanted. Next 7–8 mL of room temperature 48% HF was added and stirred. The samples were left in HF for 2 days and were stirred once per day. After centrifugation, the HF was decanted, and 7–8 mL of room temperature 10% HCl was added for 5–10 min to dissolve any precipitated fluorosilicates and subsequently rinsed twice with distilled water. The use of pH strips is recommended to ensure the sample is no longer acidic to minimize the risk of exposure to weak HF. Additional water rinses may be required. The remaining residue was then sieved a second time using only the 15 µm sieve and sonicated for 30 s. Finally, the residue was transferred to 1.5 mL vials. Fig. 1 summarizes these steps.

Slides were strew mounted on a hot plate. Glass microscope slides were placed onto the hot plate and labeled. A vortex mixer was used to homogenize the residue, and using a pipette, one to two drops of the residue were placed onto the microscope slide. The water was allowed to partially evaporate and glycerin jelly was subsequently added. A toothpick was used to homogenize the residue and glycerin jelly, and left on the hotplate for less than one minute to evaporate any remaining water. A cover slip was then applied. Dinoflagellate cysts were all counted by the same individual (P.R.G.) to circumvent observer bias, using a Zeiss Axio Imager A1 light transmitting microscope at 400× magnification. A target of 300 cysts were counted per sample in order to obtain representative relative and absolute abundances (Mertens et al., 2009).

2.1.2. Documenting loss of *Lycopodium* spores and cysts during processing

Mertens et al. (2012) report loss of *Lycopodium* spores during processing, as cyst concentrations were consistently lower when the *Lycopodium* tablets were added at the end than at the beginning. However, they did not document how many *Lycopodium* spores were lost at each stage during processing. In this study, we document how many *Lycopodium* spores and cysts are lost during decanting, discarded with the coarse (>120 µm) fraction, stuck to wooden stir sticks, lost through the fine sieve or remain stuck to sieves.

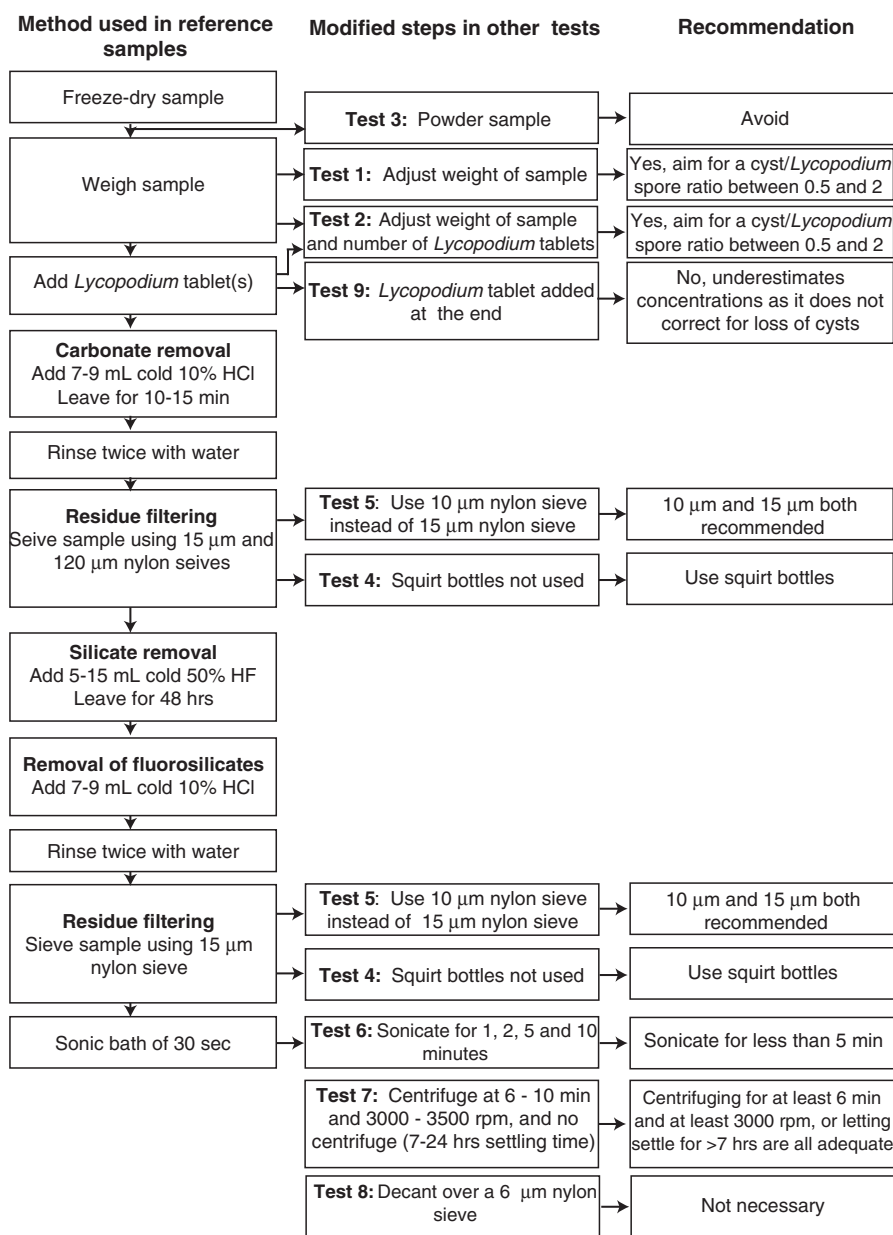


Fig. 1. Flowchart showing the method used in the reference samples (first column), and the steps modified in the other tests (second column). Recommendations are shown in the third column.

In reference samples 1–3, the number of *Lycopodium* spores and cysts were counted microscopically on a 2 cm² section of the nylon sieve. These values were used to calculate their numbers over the entire sieve area (24 cm²).

For reference samples 4–6, we captured what was lost through the 15 µm sieve by placing a 6 µm nylon sieve beneath the 15 µm sieve. The material captured on the 6 µm sieve was subsequently transferred to a new 15 mL centrifuge tube. This resulted in a total of 5–6 mL of water-residue slurry. The precise volume was recorded. The slurry was homogenized by using a vortex mixer and immediately five drops were subsampled. The number of *Lycopodium* spores and cysts in the five drops was analyzed microscopically and used to calculate the total number. The material remaining on the 120 µm sieve, the coarse fraction, was retained and also analyzed microscopically.

In reference samples 7–9, the distilled water and the HCl were decanted over a 6 µm sieve, after centrifuging. The 6 µm sieve was then analyzed for cysts and *Lycopodium* spores. Due to the extreme toxicity of HF, decanting of HF was not done over the sieve.

2.1.3. Test 1: varying the dinoflagellate cyst/*Lycopodium* spore ratio through adjustment of the sample weight

A cyst/*Lycopodium* spore ratio of two is suggested by Maher (1981) to facilitate counting; however, this ratio has not been experimentally verified. Mertens et al. (2012) found that many samples from one out of the three studied stations (station C) did not have reproducible concentrations and suggested it was due to the addition of too few *Lycopodium* tablets, resulting in a high cyst/*Lycopodium* spore ratio (average of 8.45). Usually, the dinoflagellate cyst concentrations are not known prior to palynological processing, requiring the analyst to estimate the appropriate number of tablets that need to be added. As the tablets contain a fixed number of spores, adding too few or too many tablets to a sediment sample may lead to ratios much greater than two or much smaller. To test whether samples were reproducible at different ratios, three replicates were processed using 0.6, 1.2, 2.4, 4.8, and 9.6 g of freeze-dried sediment. These weights were chosen because a preliminary analysis of a few samples suggested that these weights would result in cyst/*Lycopodium* spore ratios of 0.25, 0.5, 1,

2, and 4, respectively. Only one *Lycopodium* tablet was added to each sample prior to processing, keeping the amount of *Lycopodium* spores constant. The adjustments of the weight will increase the amount of dinoflagellate cysts counted per *Lycopodium* spores counted and will thus increase the ratio, but not the concentrations (Fig. 2). Based on previously published results (Maher, 1981; Mertens et al., 2012), we hypothesize that samples with dinoflagellate cyst/*Lycopodium* spore ratios greater than two will lead to less reproducible cyst concentrations.

2.1.4. Test 2: varying the sample size and number of *Lycopodium* tablets to keep the dinoflagellate cyst/*Lycopodium* spore ratio constant

In test 2, the mass of the sediment varied (1 g, 3 g and 5 g), as well as the number of *Lycopodium* tablets. One tablet was added for every gram of sediment (i.e., 3 g and 3 tablets), thereby keeping the ratio constant, as well as the concentrations. We hypothesize that more error is

introduced when using small amounts of sediments compared to larger amounts, as any residue lost during processing (e.g., on stir sticks, through sieving, decanting, etc.) is likely to be proportionally larger for small samples. Thus, we expect higher standard deviations of cyst concentrations in the 1 g samples compared to the 5 g samples.

2.1.5. Test 3: effect of powdering samples

Sediment that has been powdered (ground up) could be suggested to be unacceptable for palynological analysis as powdering is thought to fragment palynomorphs. Many geochemical analyses of sediments require the sediments be freeze-dried and powdered prior to analysis (e.g., CaCO_3 content—Müller and Gastner, 1971; total organic content—Bernard et al., 1995; x-ray fluorescence—Hennekam and de Lange, 2012). The purpose of test 3 is to determine if powdered samples are acceptable for determination of concentrations of dinoflagellate cysts. A mortar and pestle were used to grind the dry sediments

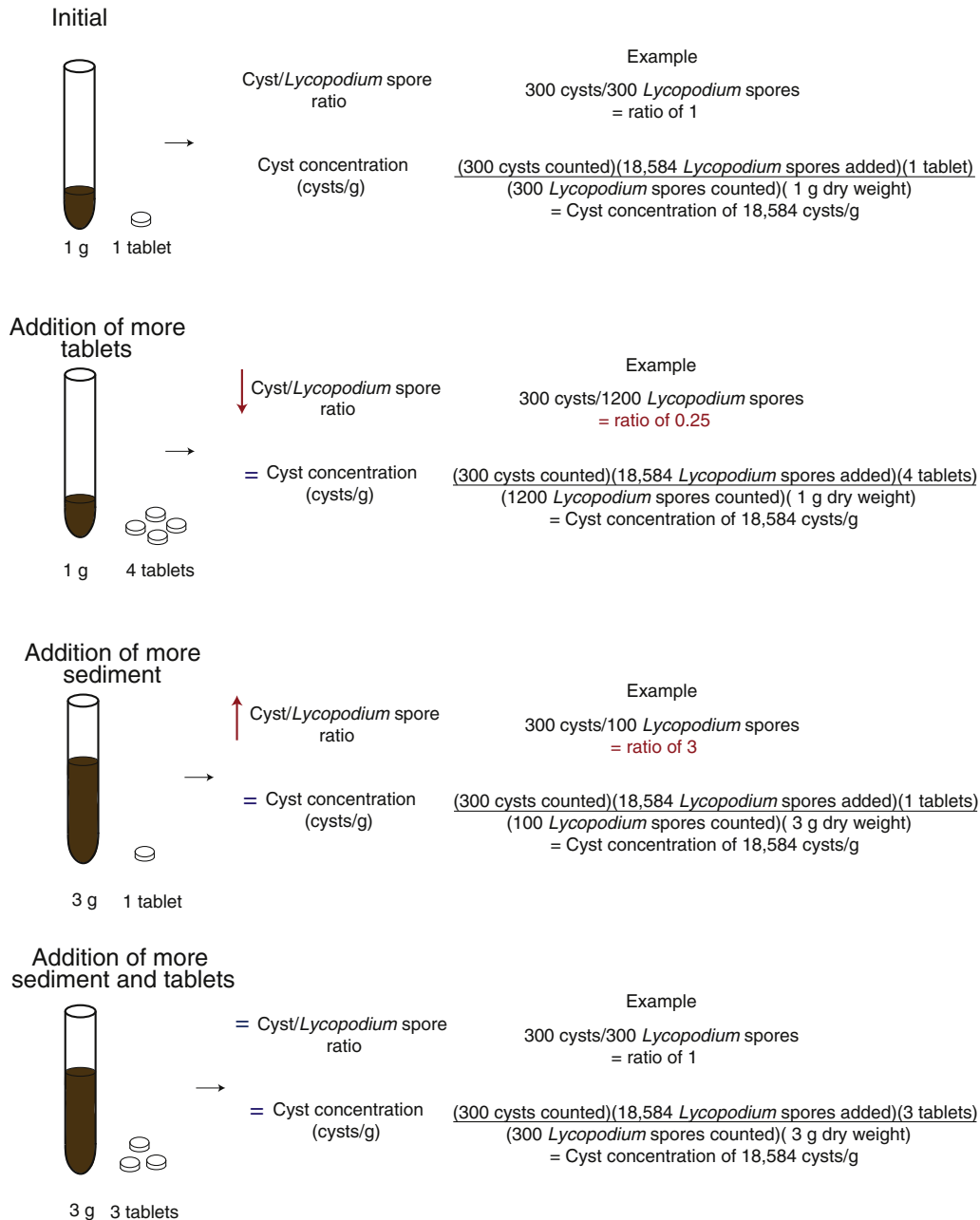


Fig. 2. Diagram showing how different sediment weights and number of *Lycopodium* tablets influence the cyst/*Lycopodium* spore ratio and total cyst concentrations.

until they were a fine powder. Subsamples of this powder were then transferred to the centrifuge tubes and weighed.

2.1.6. Test 4: No squirt bottles during sieving

The use of squirt bottles during sieving speeds up the process by pushing water through the sieve at a faster rate and by preventing the sediment from settling and clogging the sieve. Squirt bottles with large openings (~3 mm) were used during sieving, while squirt bottles with narrower openings (0.5 mm) were used to transfer the sediment from the centrifuge tubes to the sieve. To determine if using squirt bottles during sieving increases the loss of *Lycopodium* spores, we did not use squirt bottles for this test. Instead, distilled water was added during sieving using a water jug with a wide pour spout.

2.1.7. Test 5: 10 µm nylon mesh sieve

L. clavatum spores range from 20 to 40 µm in diameter, with 75% of spores ranging from 28 to 33 µm (Lignum et al., 2008), and thus should not pass through a 15 µm mesh sieve. However, Lignum et al. (2008) suggest a loss of 5.8% for 15 µm nylon meshes, which is probably due to fragmentation of the spores. To determine if fewer *Lycopodium* spores are lost by using a finer mesh, in test 5 we use 10 µm nylon sieves instead of the 15 µm nylon sieves.

2.1.8. Test 6: varying sonicating time

Mertens et al. (2009) recommend a limit of 1 min for sonicating, after documenting treatments that sonicated samples for 30 min caused extensive damage to cysts and other palynomorphs. In order to determine if 1 min is indeed, an acceptable limit samples were sonicated using a BransonT 3510R-MT ultrasonic cleaner (frequency: 42 kHz ± 6%; 100 W) for 30 s (reference set), 1 min (test 6a), 2 min (test 6b), 5 min (test 6c), and 10 min (test 6d) during the second sieving. The residue was poured into a 150 mL glass beaker and sonicated in the beaker with approximately 20 mL of water. After sonication, the residue was poured back over the 15 µm sieve and sieved longer to rinse the sample of the small particles created by sonicating.

2.1.9. Test 7: varying centrifuge time and speed

To determine if centrifuging for longer or at higher speeds retains more palynomorphs, we centrifuged for 10 min at 3000 rpm in test 7a, for 6 min at 3500 rpm in test 7b, and for 10 min at 3500 rpm in test 7c. These changes in centrifuging time and/or speeds were performed whenever the centrifuge (radius 15 cm) was used. In test 7d, no centrifuge was used; instead, these samples rested for 7–24 h, allowing the residue to settle before decanting. All centrifuge tubes were 15 mL.

2.1.10. Test 8: effect of decanting over a 6 µm sieve

In this test, we decanted the liquids (HCl and water) over a 6 µm sieve and rinsed the sieve back into the centrifuge tube. This was done at every step decanting occurred, except for HF due to the harmful nature and extreme toxicity of this acid. This method prevents cysts and *Lycopodium* spores from being lost during decanting.

2.1.11. Test 9: *Lycopodium* tablets added at the end

Mertens et al. (2012) found that adding the *Lycopodium* tablets at the end of processing resulted in significantly lower cyst concentrations, in contrast to adding them at the beginning. As none, or very few, *Lycopodium* spores can be lost when the tablets are added at the end, this suggests that some *Lycopodium* spores are lost during processing when the tablet is added at the beginning. We repeat the test in this study at a lower cyst/*Lycopodium* spore ratio.

2.2. Calculating dinoflagellate cyst abundances and statistical analyses

The relative and absolute abundance (often called concentration) of dinoflagellate cysts was calculated for each sample. Cyst

concentrations were calculated using the equation of Benninghoff (1962).

$$c = \frac{d_c \times L_t \times t}{L_c \times w}$$

where

c concentration = number of dinoflagellate cysts/g dried sediment

d_c = number of dinoflagellate cysts counted

L_t = number of *Lycopodium* spores/tablet

t = number of tablets added to the sample

L_c = number of *Lycopodium* spores counted

w = weight of dried sample (g)

In order to determine how different methodologies impact the reproducibility of dinoflagellate cyst abundances and composition, each test containing three replicates was compared to the first three samples in the reference set using Maher's test of reproducibility (Maher, 1981). Maher's test of reproducibility is an algorithm, which calculates confidence limits on microfossil concentrations and can be used to determine if microfossil concentrations from two samples are statistically similar (Maher, 1981). As in Mertens et al. (2009, 2012), a change in the algorithm was made where sediment weights were used instead of sediment volume. A sample is deemed statistically similar to a reference sample when it falls within the confidence limits (accept) but is rejected when it falls outside of the confidence limits. The first three samples from the reference set were used to compute Maher's reproducibility test on tests 1–9. Three replicates were performed for each test. For each test, the mean and standard deviations were also calculated for total cyst concentrations and cyst/*Lycopodium* spore ratios (Tables 1 and 2).

While Maher's reproducibility test is used for total cyst concentrations, other statistical tests should be used on other values (taxa concentrations, ratios, relative abundances, etc.) and are therefore used here. In order to determine if the mean values of these parameters in each test are statistically different from the mean of the reference set, one-way post hoc ANOVA (analysis of variance) tests were performed using IBM SPSS Statistics. In order to determine the appropriate post hoc to use, the Levene test (Levene, 1960) was calculated. If the significance value was greater than 0.05, it passed the test of homogeneity of variance. In this case, the Gabriel method was chosen, as it is an equal variance post hoc method that is appropriate when groups are of different sizes. If the Levene statistic had a significance value of less than 0.05, it failed the test of homogeneity of variance. In this case, the Games–Howell method was chosen as it is a post hoc method that does not assume equal variances, and can also be used when group sizes are unequal. This test also incorporates the samples sizes and variances of each group being compared.

For test 1, where the average dry weight varied (0.6, 1.2, 2.4, 4.8, 9.6 g), a linear regression was performed on the dry weights and cyst/*Lycopodium* spore ratio.

3. Results and discussion

Between 258 and 320 dinoflagellate cysts were counted per sample. Over 25 taxa were encountered (Supplementary Table 1). The assemblage is dominated by cysts produced by heterotrophic taxa, specifically round brown cysts (*Brigantedinium* spp. (avg. 5.1%), *Dubridinium* spp. (avg. 1.8%), other diplopsalids (avg. 0.6%) and other round browns (avg. 19.1%), cysts of *Protoperidinium americanum* (avg. 9.7%), spiny brown cysts (avg. 6.1%), *Quinquecuspis concreta* (avg. 5.9%), and cysts of *Archaeperidinium saanichi* (avg. 3.3%). Cysts produced by autotrophic taxa are less abundant but are predominately *Spiniferites* spp. (avg. 22.4%), *Operculodinium centrocarpum* sensu Wall and Dale (1966) (avg. 5.2%), cyst of *Pentaparsodinium dalei* (avg. 4.2%), and cyst of

Table 1
Results from test 1. Number of *Lycopodium* spores counted, cyst/*Lycopodium* spore ratios, total cyst concentrations, average cyst concentrations and the standard deviation, average cyst/*Lycopodium* spore ratios and the standard deviation, Maher's test when compared to three reference samples, and UVic ID number.

Test no.	Method used (1 <i>Lycopodium</i> tablet)	Cysts counted	<i>Lycopodium</i> spores counted	Cyst/ <i>Lycopodium</i> ratio	Concentration (cysts/g)	Average concentration (cysts/g)	Standard deviation	Average Cyst/ <i>Lycopodium</i> ratio	Standard deviation	Maher's test of reproducibility				Uvic ID no.
										Ref 1 (11815)	Ref 2 (10143)	Ref 3 (10830)		
1a	Dry weight 0.6 g (rep 1)	294	1034	0.28	8,720					Accept	Accept	Accept	13-338	
1a	Dry weight 0.6 g (rep 2)	302	971	0.31	9,570					Accept	Accept	Accept	13-339	
1a	Dry weight 0.6 g (rep 3)	302	995	0.30	9,323	9,204	437	0.30	0.01	Accept	Accept	Accept	13-340	
1b	Dry weight 1.2 g (rep 1)	299	425	0.70	10,832					Accept	Accept	Accept	13-326	
1b	Dry weight 1.2 g (rep 2)	308	383	0.80	12,402					Accept	Accept	Accept	13-327	
1b	Dry weight 1.2 g (rep 3)	291	426	0.68	10,518	11,251	1,010	0.73	0.06	Accept	Accept	Accept	13-328	
1c	Dry weight 2.4 g (rep 1)	297	244	1.22	9,351					Accept	Accept	Accept	13-323	
1c	Dry weight 2.4 g (rep 2)	307	210	1.46	11,231					Accept	Accept	Accept	13-324	
1c	Dry weight 2.4 g (rep 3)	296	166	1.78	13,710	11,431	2,186	1.49	0.28	Accept	Accept	Accept	13-325	
1d	Dry weight 4.8 g (rep 1)	304	92	3.30	12,738					Accept	Accept	Accept	13-272	
1d	Dry weight 4.8 g (rep 2)	309	103	3.00	11,567					Accept	Accept	Accept	13-273	
1d	Dry weight 4.8 g (rep 3)	301	94	3.20	12,336	12,213	595	3.17	0.15	Accept	Accept	Accept	13-274	
1e	Dry weight 9.6 g (rep 1)	291	38	7.66	14,737					Accept	Accept	Accept	13-275	
1e	Dry weight 9.6 g (rep 2)	311	49	6.35	12,219					Accept	Accept	Accept	13-276	
1e	Dry weight 9.6 g (rep 3)	286	29	9.86	18,983	15,313	3,418	7.96	1.78	Accept	Reject	Reject	13-277	

cf. *Alexandrium* spp. (avg. 3.0%). Cysts of *Biecheleria* sp. were also counted, but due to their small size (~10–30 µm), where some may get lost during sieving, they were excluded.

The nine reference samples had total cyst concentrations ranging from 9106 to 11,815 cysts/g, with an average of 10,691 cysts/g and a standard deviation of 753 cysts/g. Given the range of concentrations, it suggests a random error of approximately 10–15% in absolute cyst abundances. The largest variation (maximum % – minimum %) in these samples occurs for *Brigantedinium* spp. (10.6%), RBCs (10.6%), cysts of *P. americanum* (5.6%), cyst of cf. *Alexandrium* spp. (4.8%), *Spiniferites* spp. (4.7%), cyst of *P. dalei* (4.6%), and *Q. concreta* (4.2%), with standard deviations between 1.4 and 3.7. Taxa with average percentages below 5% displayed variations of 0.3 to 4.7% and standard deviations of 0.1 to 1.4%. This suggests that random variation ranges from approximately 4 to 11% in the most dominant taxa and approximately 0.5 to 5% in rarer taxa. There were 15 taxa that were recorded in at least one of the reference samples, but that were not recorded in all the reference samples. Species richness in the reference set ranges from 22 to 26.

As some processing techniques are known to destroy sensitive cysts, we compared cyst assemblages across methodologies. Cysts were grouped into three categories based on their resistance to degradation, following Zonneveld et al. (2001) and Mertens et al. (2012).

Extremely sensitive cysts: *Brigantedinium cariacense*, *Brigantedinium simplex*, *Brigantedinium* spp., cyst of *Alexandrium* spp., cyst of *Polykrikos kofoidii* sensu Matsuoka et al. (2009), cyst of *Polykrikos schwartzii* sensu Matsuoka et al. (2009), cyst of *P. americanum*, cyst of *Protoperidinium nudum*, cyst of *A. saanichi*, cyst of *Archaeoperidinium minutum*, *Dubridinium* spp., *Echinidinium aculeatum*, *Echinidinium delicatum*, *Echinidinium granulatatum*, *Echinidinium transparentum*, *Protoperidinium* spp., round brown cysts (RBC), spiny round brown cysts (SBC), *Selenopemphix quanta*, *Selenopemphix undulata*, and *Trinovantedinium variabile*.

Moderately sensitive cysts: *O. centrocarpum* sensu Wall and Dale (1966), *Q. concreta*, *Spiniferites bentorii*, *Spiniferites elongatus*, *Spiniferites membranaceus*, *Spiniferites ramosus*, *Spiniferites* spp., *Votadinium calvum*, and *Votadinium spinosum*.

Resistant cysts: cyst of *P. dalei*.

The average of extremely sensitive cysts for the nine reference samples is 60.5%, for moderately sensitive 35.3%, and only 4.2% were resistant cysts.

3.1. Documenting loss during processing of the reference samples

3.1.1. Loss on sticks and sieves

In the first three reference samples, we analyzed the residue stuck on the sieves and wooden disposable stir sticks after they were finished with. In the first sieving between 202 and 238, *Lycopodium* spores were documented on the 15 µm sieve, and between 24 and 36 dinoflagellate cysts. During the second sieving between 107 and 154, *Lycopodium* spores were documented on the 15 µm sieve, and between 24 and 144 cysts (Table 4). A total of six stir sticks were used on each sample and analyzed (first—HCl, second after the *Lycopodium* tablet has been added, third—water rinse, fourth—water rinse, fifth—HCl, sixth—water rinse). Additional stir sticks were used to stir the samples after HF was added, but for safety reasons, these were not analyzed. For all samples, we observed only minerals and amorphous organic material on the first three sticks, with the exception of sample number 13–278 where two *Lycopodium* spores were also observed on one of the sticks. In the later three sticks, an average of one *Lycopodium* spore and one cyst were found on each stick (Table 4).

Table 2
Cyst/*Lycopodium* spore ratios, total cyst concentrations, average cyst concentrations and the standard deviation, average cyst/*Lycopodium* spore ratios and the standard deviation, Maher's test when compared to three reference samples, and UVic ID number.

Test no.	Method used	Cyst/ <i>Lycopodium</i> ratio	Concentration (cysts/g)	Average concentration	Standard deviation	Average cyst/ <i>Lycopodium</i> ratio	Standard deviation	Maher's test of reproducibility			Uvic ID no.
								Ref 1 (11815)	Ref 2 (10143)	Ref 3 (10830)	
	Reference (ref 1)	0.64	11,815	10,691	753	0.58	0.04		Accept	Accept	13–278
	Reference (ref 2)	0.55	10,143					Accept		Accept	13–279
	Reference (ref 3)	0.59	10,830					Accept	Accept		13–280
	Reference (ref 4) + document loss through sieves	0.60	10,989					Accept	Accept	Accept	13–287
	Reference (ref 5) + document loss through sieves	0.59	10,966					Accept	Accept	Accept	13–288
	Reference (ref 6) + document loss through sieves	0.49	9,106					Accept	Accept	Accept	13–289
	Reference (ref 7) + document loss during decanting	0.59	10,852					Accept	Accept	Accept	13–290
	Reference (ref 8) + document loss during decanting	0.60	11,103					Accept	Accept	Accept	13–291
	Reference (ref 9) + document loss during decanting	0.57	10,413					Accept	Accept	Accept	13–292
2a	3 g and 3 <i>Lycopodium</i> tablets (rep 1)	0.70	12,929	11,906	1,037	0.64	0.06	Accept	Accept	Accept	13–281
2a	3 g and 3 <i>Lycopodium</i> tablets (rep 2)	0.64	11,934					Accept	Accept	Accept	13–282
2a	3 g and 3 <i>Lycopodium</i> tablets (rep 3)	0.58	10,856					Accept	Accept	Accept	13–283
2b	5 g and 5 <i>Lycopodium</i> tablets (rep 1)	0.64	11,961	10,818	1,273	0.58	0.07	Accept	Accept	Accept	13–284
2b	5 g and 5 <i>Lycopodium</i> tablets (rep 2)	0.51	9,445					Accept	Accept	Accept	13–285
2b	5 g and 5 <i>Lycopodium</i> tablets (rep 3)	0.59	11,047					Accept	Accept	Accept	13–286
3	Powdered (rep 1)	0.48	8,866	8,670	205	0.47	0.01	Accept	Accept	Accept	13–317
3	Powdered (rep 2)	0.46	8,457					Accept	Accept	Accept	13–318
3	Powdered (rep 3)	0.47	8,688					Accept	Accept	Accept	13–319
4	No squirt bottles (rep 1)	0.59	10,889	11,793	2,388	0.64	0.13	Accept	Accept	Accept	13–329
4	No squirt bottles (rep 2)	0.54	9,989					Accept	Accept	Accept	13–330
4	No squirt bottles (rep 3)	0.78	14,501					Accept	Reject	Accept	13–331
5	10 μ mmesh (rep 1)	0.61	11,330	10,905	732	0.59	0.04	Accept	Accept	Accept	13–320
5	10 μ mmesh (rep 2)	0.61	11,325					Accept	Accept	Accept	13–321
5	10 μ mmesh (rep 3)	0.54	10,059					Accept	Accept	Accept	13–322
6a	Sonicate 1 min (rep 1)	0.73	13,698	11,538	1,899	0.62	0.10	Accept	Accept	Accept	13–305
6a	Sonicate 1 min (rep 2)	0.55	10,130					Accept	Accept	Accept	13–306
6a	Sonicate 1 min (rep 3)	0.58	10,786					Accept	Accept	Accept	13–307
6b	Sonicate 2 min (rep 1)	0.46	8,608	10,240	1,858	0.55	0.10	Accept	Accept	Accept	13–308
6b	Sonicate 2 min (rep 2)	0.53	9,851					Accept	Accept	Accept	13–309
6b	Sonicate 2 min (rep 3)	0.66	12,262					Accept	Accept	Accept	13–310
6c	Sonicate 5 min (rep 1)	0.60	11,177	10,251	2,129	0.55	0.12	Accept	Accept	Accept	13–311
6c	Sonicate 5 min (rep 2)	0.42	7,815					Reject	Accept	Accept	13–312
6c	Sonicate 5 min (rep 3)	0.64	11,760					Accept	Accept	Accept	13–313
6d	Sonicate 10 min (rep 1)	0.63	11,680	13,721	1,774	0.74	0.10	Accept	Accept	Accept	13–314
6d	Sonicate 10 min (rep 2)	0.80	14,882					Accept	Reject	Accept	13–315
6d	Sonicate 10 min (rep 3)	0.79	14,602					Accept	Reject	Accept	13–316
7a	Centrifuge 3,000 rpm for 10 min (rep1)	0.75	13,875	12,623	1,096	0.68	0.06	Accept	Accept	Accept	13–293
7a	Centrifuge 3,000 rpm for 10 min (rep2)	0.64	11,834					Accept	Accept	Accept	13–294
7a	Centrifuge 3,000 rpm for 10 min (rep3)	0.65	12,160					Accept	Accept	Accept	13–295
7b	Centrifuge 3,500 rpm for 6 min (rep 1)	0.61	11,225	11,841	1,342	0.64	0.07	Accept	Accept	Accept	13–296
7b	Centrifuge 3,500 rpm for	0.59	10,917					Accept	Accept	Accept	13–297

Table 2 (continued)

Test no.	Method used	Cyst/ <i>Lycopodium</i> ratio	Concentration (cysts/g)	Average concentration	Standard deviation	Average cyst/ <i>Lycopodium</i> ratio	Standard deviation	Maher's test of reproducibility			Uvic ID no.
								Ref 1 (11815)	Ref 2 (10143)	Ref 3 (10830)	
7b	6 min (rep 2) Centrifuge 3,500 rpm for 6 min (rep 3)	0.72	13,380					Accept	Accept	Accept	13–298
7c	Centrifuge 3,500 rpm for 10 min (rep 1)	0.54	9,964	9,916	1,121	0.54	0.06	Accept	Accept	Accept	13–299
7c	Centrifuge 3,500 rpm for 10 min (rep 2)	0.47	8,772					Accept	Accept	Accept	13–300
7c	Centrifuge 3,500 rpm for 10 min (rep 3)	0.60	11,013					Accept	Accept	Accept	13–301
7d	No centrifuge (rep 1)	0.68	12,570	12,265	811	0.66	0.04	Accept	Accept	Accept	13–302
7d	No centrifuge (rep 2)	0.69	12,880					Accept	Accept	Accept	13–303
7d	No centrifuge (rep 3)	0.61	11,346					Accept	Accept	Accept	13–304
8	Decant over 6 μ m sieve (rep 1)	0.65	12,146	12,155	887	0.65	0.05	Accept	Accept	Accept	13–332
8	Decant over 6 μ m sieve (rep 2)	0.70	13,047					Accept	Accept	Accept	13–333
8	Decant over 6 μ m sieve (rep 3)	0.61	11,273					Accept	Accept	Accept	13–334
9	<i>Lycopodium</i> tablet at end (rep 1)	0.45	8,254	8,007	525	0.43	0.03	Reject	Accept	Accept	13–335
9	<i>Lycopodium</i> tablet at end (rep 2)	0.45	8,362					Reject	Accept	Accept	13–336
9	<i>Lycopodium</i> tablet at end (rep 3)	0.40	7,404					Reject	Accept	Reject	13–337

3.1.2. Loss through the 15 μ m and on 120 μ m sieves

For reference samples 4–6, we captured and documented the number of *Lycopodium* spores and cysts lost through the 15 μ m sieve and those that remained in the coarse fraction. During the first sieving between 418 and 734, *Lycopodium* spores were lost, and 70 to 108 cysts. During the second sieving, 400 to 1206 *Lycopodium* spores were lost, and 90 to 220 cysts (Table 4). Less than 20 *Lycopodium* spores and cysts were found in the coarse fraction.

3.1.3. Loss during decanting

For reference samples 7–9, we analyzed what was lost during decanting. Loss of *Lycopodium* spores and cysts was low (3–9 *Lycopodium* spores per sample and 8–17 cysts per sample) (Table 4), and only occurred during the first HCl treatment, following the two water rinses, and before transferring the organic residue to 1.5 mL vials.

3.1.4. Total loss of palynomorphs during processing

The greatest loss of *Lycopodium* spores and cysts occurs during the sieving process, particularly through the 15 μ m Nylon Nitex sieves. Some remain stuck to the nylon sieves, or are retained in the coarse fraction, but the largest loss is through the sieves (Table 4). Negligible amounts are lost on the stir sticks and through decanting. Approximately four times the amount of *Lycopodium* spores during sieving are lost compared to cysts. This may be a result of the different size ranges of dinoflagellate cyst taxa, as many of the cysts are larger than the 20–40 μ m *Lycopodium* spores, and thus less likely to be lost through the fine sieve. This disproportionate loss leads to inflated calculated cyst concentrations. While only sieving once should hypothetically result in less loss of both cysts and *Lycopodium* spores, pre-sieving is often necessary when the sediments are coarser than silt, or if the sample has a high percentage of clay-silt sized silicate minerals or diatoms that react vigorously in HF. If all samples are sieved in the same manner, the cyst concentration inflation due to sieving should be consistent, and thus comparable.

The total average number of *Lycopodium* spores and cysts lost during processing (through the 15 μ m sieves, on the sieves and stir sticks, in the coarse fraction, through decanting) is 1523 or 8.2% of the total number of *Lycopodium* spores added per sample and 364 dinoflagellate cysts or 3.4% of the calculated cyst concentration, respectively (Table 4).

Taking both the loss of *Lycopodium* spores and cysts into account, we calculated a new average cyst concentration of the reference set using the Beninghoff equation by subtracting 1523 *Lycopodium* spores from the 18,584 added, and then adding 364 cysts to this calculated concentration value. This calculation results in a new calculated cyst concentration of 10,179 cysts/g. This suggests the average concentration of 10,691 cysts/g (Table 2) is overestimated by only 4.8%.

3.2. Test 1: varying the cyst/*Lycopodium* spore ratio

Five different dry weights were used in test 1 (0.6, 1.2, 2.4, 4.8, and 9.6 g), while only one *Lycopodium* tablet was added to each of these samples. When the cyst/*Lycopodium* spore ratios are plotted against dry weights, the regression slope should be one, as doubling the weight of sample should result in twice the number of cysts counted per *Lycopodium* spore counted. However, the regression between dry weight and the cyst/*Lycopodium* spore ratio yielded a slope of less than one. When no y-intercept is specified, the slope is 0.84 with an R^2 value of 0.94 ($p < 0.001$), or 0.78 with an R^2 value of 0.93 when the y-intercept is set to zero (Fig. 2a). A y-intercept of zero is perhaps most realistic as a weight of zero (or no sample) would have zero cysts. The number of cysts counted is lower than expected, and the difference becomes greater with heavier weights. One possible explanation for this result is that fewer *Lycopodium* spores were encountered in the 9.6 g samples (average 39) compared to the 0.6 g samples (average 1000), thus leading to more potential errors in the calculating the cyst/*Lycopodium* spore ratio as the coefficient of variation changes from 0.259 in the 9.6 g sample to 0.032 in the 0.6 g sample. Furthermore, based on the number of *Lycopodium* spores counted, only ~0.2% of 9.6 g samples was analyzed compared to ~5.4% of the 0.6 g samples.

The standard deviation in the cyst/*Lycopodium* spore ratio is generally larger as this ratio increases and ranges from 0.1 when the average ratio is 0.3 (0.6 g of sediment) to 1.78 when the average ratio is 7.96 (9.6 g of sediment) (Table 1, Fig. 3d). A similar trend is observed in the standard deviation calculated from the cyst concentration of the three replicates of each test, with the exception of samples with a weight of 4.8 g. The samples with 0.6 g of sediment have the smallest cyst concentration standard deviation at 437 cysts/g, and samples with 9.6 g of

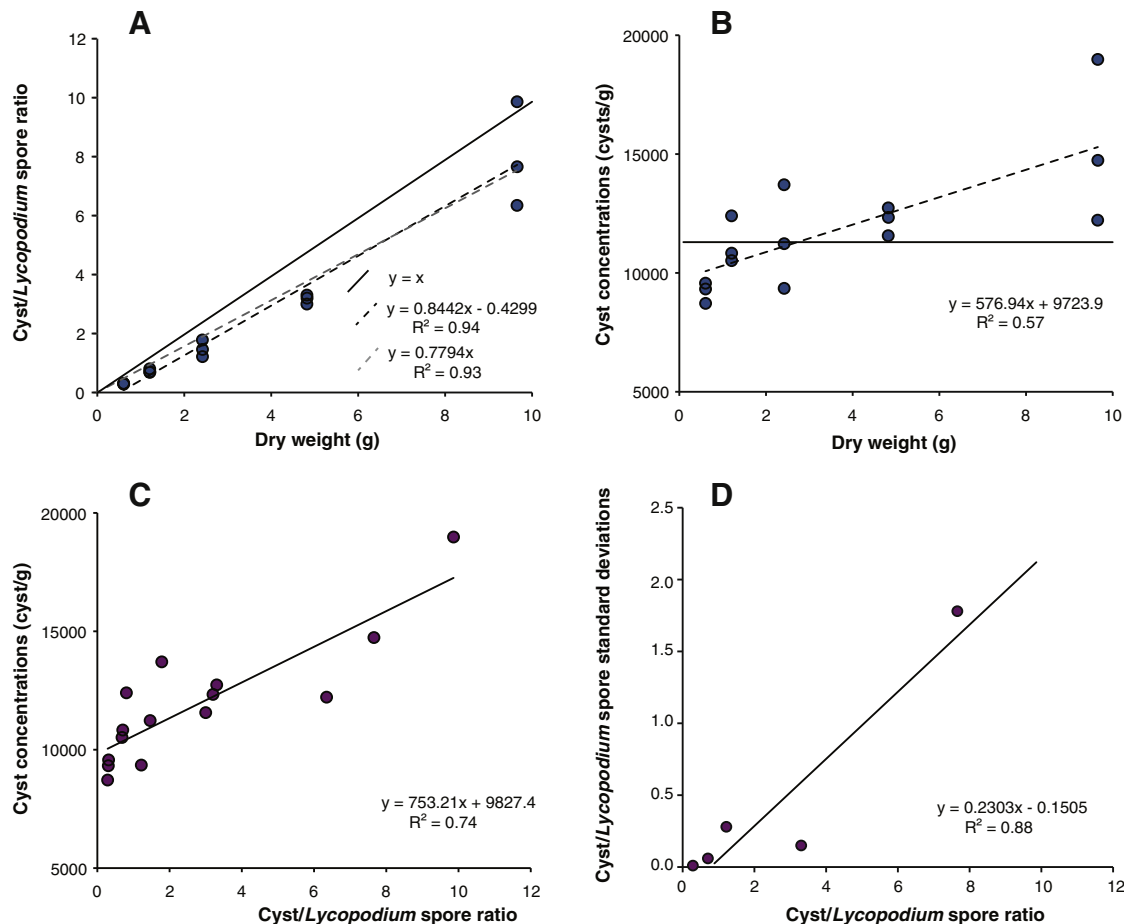


Fig. 3. (A) Cyst/Lycopodium spore ratio versus dry weight. The solid line shows a 1:1 relationship. The black dashed line shows the trend line through the data points when the y-intercept is not specified. The grey dashed line shows the trend line when the y-intercept is set to zero. (B) Cyst concentrations versus dry weight. The solid line shows the average cyst concentration when the cyst/Lycopodium spore ratio is ~2. The dashed line shows the trend line. (C) Cyst concentrations versus cyst/Lycopodium spore ratio. (D) Cyst/Lycopodium spore standard deviations versus the cyst/Lycopodium spore ratios.

sediment have the largest cyst concentration standard deviation at 3418 cysts/g (Table 1).

In theory, cyst concentrations should not depend on the weight of the sample, as doubling the mass of a sample should also lead to a doubling in the number of cysts, but keep the concentrations constant (cysts per gram sediment) (see Fig. 2). This however is not observed in this study. Instead, there is a general positive trend in total cyst concentrations with increasing dry weight and increasing cyst/Lycopodium spore ratios (Fig. 3b and c). Calculated total cyst concentrations rise (+576 cysts for each gram of sediment added) as the sample weight increased (Fig. 3b). When the number of *Lycopodium* tablets is not increased with sediment weight, fewer *Lycopodium* spores are encountered in the slides (Table 1), leading to higher calculated cyst concentrations. Coefficients of variation were calculated for the number of *Lycopodium* spores counted and were found to increase as the number of *Lycopodium* spores decreased, from 0.03 to 0.26 with the exception of 4.8 g samples, which has a low coefficient of variation (0.06). Despite these differences, Maher's test of reproducibility deemed all samples in this test statistically similar to the reference set, except sample 13–277, which was rejected against two of the replicates (Table 1). It is also of note that low amounts of sediment (0.6 g, test 1a) caused a significant increase in the number of *Lycopodium* spores that needed to be counted, which slows down the counting.

We suggest altering the number of *Lycopodium* tablets added to obtain cyst/Lycopodium spore ratios of less than five, ideally between 0.5 and 2 to facilitate counting. When ratios are below 0.5, large numbers of *Lycopodium* spores are encountered, which slows counting. If large

ratios are encountered and it is not possible or feasible to re-process the sample we recommend continuing the microscopic analysis until a greater number of *Lycopodium* spores and cysts are counted.

3.3. Test 2: varying the sample size and number of *Lycopodium* tablets to keep the dinoflagellate cyst/Lycopodium spore ratio constant

When the sample weight differed (1 g, 3 g, and 5 g) and the number of *Lycopodium* tablets added were adjusted in the same manner (1, 3, and 5 tablets), thus keeping the dinoflagellate cyst to *Lycopodium* spore ratio approximately constant, the cyst concentrations from the 3 g and 5 g samples were both statistically similar to the reference set with 1 g. The standard deviations on the cyst/Lycopodium spore ratio and standard deviations of the total cyst concentrations both increased with increasing sample weight (Table 2). This suggests that our hypothesis was incorrect. It does not appear that smaller samples result in higher standard deviations in total cyst concentrations, at least in the weights used in this study.

3.4. Test 3: effect of powdering sediment prior to processing

The concentrations of samples that were manually powdered were all statistically similar to the three reference samples; however, total cyst concentrations were consistently lower with an average of 2020 fewer cysts/g (–23%) (Table 3). The cyst/Lycopodium spore ratio was also only 0.47 compared to 0.58 in the reference set, and this difference is statistically significant (Table 3). When looking at the cyst assemblages,

Table 3

Mean difference between nine reference samples and the samples studied in the other tests (3 replicate samples per test). Values in bold are statistically significant ($p < 0.05$). Positive values are lower than the reference samples, while negative values are higher than the reference samples. Gab—Gabriel post hoc test. G-H—Games–Howell post hoc test.

Test no.	Total Concentrations (G-H)	Cyst/ <i>Lycopodium</i> ratio (G-H)	Total heterotrophs concentrations (Gab)	Total autotrophs concentrations (G-H)	Autotrophic/Heterotrophic ratio (G-H)	RBC concentrations (Gab)	SBC concentrations (G-H)	Peridinoid concentrations (Gab)	<i>Operculodinium centrocarpum</i> concentrations (Gab)
2a	−1,215	−0.062	−613	−615	−0.033	221	−151	−489	−341
2b	−127	−0.003	271	−228	−0.061	456	−363	−120	−110
3	2,020	0.110	2,363	−326	−0.347	1,528	131	223	−43
4	−1,102	−0.057	−849	−266	0.034	−288	−265	−121	−49
5	−214	−0.009	−367	140	0.048	220	−611	−167	−85
6a	−847	−0.041	−776	−84	0.049	−205	−635	237	57
6b	451	0.028	630	−192	−0.073	486	−268	510	−22
6c	440	0.025	525	−60	−0.044	515	164	337	−60
6d	−3,031	−0.159	−281	−3,083	−0.407	1,144	−183	−61	−746
7a	−1,932	−0.101	−1,261	−684	0.005	−940	80	−274	−119
7b	−1,150	−0.060	−633	−529	−0.022	−235	−495	−57	−249
7c	774	0.043	−33	805	0.126	−270	−121	155	256
7d	−1,574	−0.081	−1,437	−150	0.078	−940	−387	−118	−250
8	−1,465	−0.076	−1,017	−447	0.018	−540	−191	−295	74
9	2,684	0.147	1,678	994	0.023	651	423	371	36

there are a statistically significant fewer number of cysts produced by heterotrophic taxa in the samples; a mean of 4574 heterotrophic cysts/g in the powdered samples compared to 6938 heterotrophic cysts/g in the reference set. This difference is mainly caused by fewer RBCs. In contrast, concentrations of cysts produced by autotrophic taxa are similar to that of the reference set. This results in a statistically significant difference in the autotrophic/heterotrophic ratio (Table 3), with an average of 0.89 for the powdered samples and 0.54 for the reference set. Samples that were powdered (test 3) also have lower relative abundances of extremely sensitive cysts, 53%, but this mean value is not statistically significantly different from the mean of the reference samples according to a one-way Gabriel post hoc ANOVA test.

The grinding of these samples likely resulted in the cysts and other palynomorphs becoming broken and fragmented. The fragments are then more likely to pass through the 15 μ m sieve. Mechanical damage was not noticeably greater in samples from this test; however, cysts produced by heterotrophic dinoflagellates are more prone to mechanical degradation than cysts produced by autotrophic taxa (e.g., Mertens et al., 2009). This can explain why there are significantly fewer concentrations of cysts produced by heterotrophic taxa, whereas concentrations of cysts produced by autotrophic taxa remain unchanged. We thus recommend avoiding using sediments that have been powdered (ground up) for dinoflagellate cyst analysis.

3.5. Test 4: No squirt bottles during sieving

To determine whether using squirt bottles with narrow tips causes the loss of *Lycopodium* spores during sieving, in test 4 we used a water pitcher with a wide pour spout in place of the squirt bottles. The pitcher adds water gently with very little water pressure. Conversely, depending on the diameter of the nozzle tip, a much higher water pressure can be achieved with the squirt bottles. The higher pressure obtained by using the squirt bottle helps to rinse the samples faster and thus is more time efficient. One (13–331) of the replicate samples was not reproducible with reference sample 2 (Table 2) due to its higher total cyst concentrations. This may be because fewer cysts were lost during processing. The other two samples had similar total concentrations to the reference samples. Using pitchers of water instead of squirt bottles is not recommendable, and we recommend using squirt bottles to expedite the sieving process.

3.6. Test 5: 10 μ m mesh sieve

If a large number of *Lycopodium* spores are lost through 15 μ m sieves, presumably fewer should be lost when using a 10 μ m sieve, thus

resulting in lower total cyst concentrations. Loss of *Lycopodium* spores may not only occur through the mesh but become entrapped within the mesh itself. Samples from this test only have 2.0% more cysts than the reference samples where 15 μ m sieves were used, which is not statistically different according to Maher's test of reproducibility (Table 2) and a post hoc one-way ANOVA test. This suggests that using a 10 μ m mesh is not significantly better at retaining *Lycopodium* spores than using a 15 μ m mesh.

3.7. Test 6: varying sonicating time

The concentrations of samples that were sonicated for 1 min and 2 min are all statistically similar to the three reference samples. For samples that were sonicated for 5 min, all samples except for one were statistically similar. For samples that were sonicated for 10 min (test 6d), samples were statistically similar to only two of the reference samples and had an average of 3031 more cysts/g (+22%) than the reference samples. The average cyst/*Lycopodium* spore ratio of samples from test 6d was 0.74 compared to 0.58 of the reference samples (Table 2). Samples from test 6d had lower relative and absolute abundances of RBCs and higher abundances of *O. centrocarpum* sensu Wall and Dale (1966) and *Spiniferites* spp. (Table 3). These changes in the species composition resulted in 14.4% fewer extremely sensitive and 12.8% more moderately sensitive cysts, both of which are statistically significant (Table 3). Samples that were sonicated for 10 min (test 6d) have a much lower average value of extremely sensitive cysts at 46.9%. This difference is statistically significant according to a one-way Gabriel post hoc ANOVA test (Table 3).

Fragmented or broken cysts are more easily lost during sieving due to their smaller size. The same is true for *Lycopodium* spores. Mertens et al. (2009) document extensive damage to cysts that were sonicated for 30 min (lab 18a). They document fragmentation of cysts and report broken and loss of spines in the four samples processed using this technique. Three out of the four samples have lower cyst concentrations (Mertens et al., 2009 supplementary data), suggesting that more cysts were lost than *Lycopodium* spores. The opposite is reported in this study, as samples sonicated for 10 min had higher total cyst concentrations and higher cyst/*Lycopodium* spore ratios, suggesting instead that the long sonication time caused a preferential loss of *Lycopodium* spores. This lab also used KOH and 20 μ m sieves instead of 15 μ m sieves, which may explain the difference.

In the current study, all samples sonicated for 2 min or less were statistically similar to the three reference samples and therefore reproducible. At 5 and 10 min, cyst concentrations were less reproducible. A sonication time of 30 s is often adequate to break up clumps; however,

Table 3 (continued)

<i>Spiniferites</i> spp. concentrations (Gab)	Extremely sensitive cyst concentrations (Gab)	Moderately sensitive cyst concentrations (G-H)	Resistant cyst concentrations (Gab)	RBC cysts % (Gab)	SBC cysts % (G-H)	Peridinoid cysts % (Gab)	<i>Operculodinium</i> <i>centrocarpum</i> % (Gab)	<i>Spiniferites</i> spp. % (Gab)	Extremely sensitive cyst % (Gab)	Moderately sensitive cyst % (Gab)	Resistant cyst % (Gab)
–258	–360	–719	–122	4.75	–0.12	–3.04	–2.28	.31	3.09	–2.37	–.61
–163	295	–300	–122	4.40	–3.02	–1.06	–0.90	–1.41	3.65	–2.66	–.99
–279	1,794	183	73	10.65	–1.52	0.06	–1.62	–8.61	6.65	–6.15	–.16
–334	–739	–346	–18	–0.29	–0.39	–0.41	0.16	–.49	–.66	.42	.24
448	–481	310	–42	2.59	–5.42	–1.35	–0.70	4.54	–3.08	3.47	–.39
–127	–841	0	–6	0.02	–4.31	2.82	0.91	.60	–2.97	2.69	.28
–195	311	110	30	3.17	–3.28	4.42	–0.31	–2.61	.20	–.19	–.01
–41	164	251	64	3.50	1.63	2.40	–0.96	–1.00	–1.42	1.11	.64
–2,245	126	–2,811	–297	14.77	1.83	1.79	–4.20	–11.40	14.42	–12.80	–1.29
–369	–1,438	–454	–40	–3.08	2.69	–0.61	–0.22	.70	–2.14	1.72	.42
–310	–614	–606	70	0.64	–2.60	0.55	–1.62	–.46	.79	–1.75	.96
633	–318	1,052	50	–5.27	–2.30	0.66	2.09	4.78	–8.09	7.96	.25
2	–1,270	–292	–12	–3.85	–1.45	0.29	–1.35	2.85	–2.48	2.03	.46
–354	–787	–597	–68	–1.06	0.02	–1.21	1.30	–.09	.72	–.60	–.01
885	1,412	1,232	40	–2.08	0.86	1.03	–1.23	3.45	–2.76	3.59	–.84

if necessary, we suggest extending the sonication time to no longer than 2 min in total. Thus, we amend Mertens et al. (2009) proposed limit of 60 s to a new limit of 2 min.

3.8. Test 7: varying centrifuge times and speeds

The concentrations of samples in tests 7a–7c where the centrifuge speed varied from 3000 to 3500 rpm and where the time varied from 6 min to 10 min, and in test 7d where no centrifuge was used, were all statistically similar to the reference samples (Table 2). Samples that were centrifuged at 3500 rpm for 10 min (test 7c) had the highest relative abundance of extremely sensitive cysts at 68.6%, but again this difference was not statistically significant according to a one-way Gabriel post hoc ANOVA test (Table 3).

The samples where the residue was allowed to settle for 7–24 h before decanting, instead of using a centrifuge to concentrate the residue at the bottom of the tube, had similar total cyst concentrations to the reference samples where centrifuging was used. While letting samples settle appears to be adequate, it is more time efficient to centrifuge for 6 min. As there is no difference between 3000 and 3500 rpm we recommend using 3000 rpm as it is gentler on the centrifuge.

3.9. Test 8: decanting over a 6 µm sieve

We see a slight increase in cyst concentrations and in the cyst/*Lycopodium* spore ratio in this test, where any material lost during decanting (except HF) was captured and returned to the sample, but this increase is not statistically significant (Table 2). We suspect the small increase in cyst concentrations in this method is due to the observation that more cysts than *Lycopodium* spores are lost during

decanting (reference samples 7–9). As this method requires more time, and as there is no statistical difference, we do not recommend decanting over a 6 µm sieve.

3.10. Test 9: *Lycopodium* tablets at the end of processing

When *Lycopodium* tablets were added at the end of sample preparation Mertens et al. (2009) reported that cyst concentrations were consistently lower and suggested a loss of *Lycopodium* spores between 42% and 45% in samples where the *Lycopodium* spores were added at the beginning. In this study, we also report lower cyst concentrations, where 4 out of the 9 comparisons were found to not be statistically similar to the reference set. Concentrations were an average of 2684 cysts/g (–33.5%) lower than the three reference samples, and the average cyst/*Lycopodium* spore ratio was 0.43 (Table 2), both of which are statistically different according to Games–Howell post hoc one-way ANOVA tests (Table 3). Concentrations of SBCs and moderately sensitive cysts are statistically lower than the three reference samples (Table 3).

In Table 4, we document an average loss of only 1523 *Lycopodium* spores or 8.2% in samples where the *Lycopodium* spores were added at the beginning. There are two possible explanations for this discrepancy. First, some *Lycopodium* spores may have been missed during counting, as they can be hidden behind clumps of amorphous organic matter. Second, while Mertens et al. (2009) assumes that cyst concentrations of samples where *Lycopodium* tablets were added at the end are the “true” concentrations, this is only correct if one assumes that no cysts are lost during processing. This is not the case. Thus, adding the *Lycopodium* tablet at the end underestimates the “true” total cyst concentrations. Conversely, adding *Lycopodium* tablets at the beginning of processing overestimates cyst concentrations, as

Table 4

Documenting loss of *Lycopodium* spores and cysts on sieves, through sieves, within the coarse (>120 µm) fraction, and during decanting in the reference samples.

	Uvic ID no.	Number of <i>Lycopodium</i> spores	Number of cysts
Loss on 15 µm sieve during first and second sieving	13–278	238 + 154 = 392	36 + 48 = 84
	13–279	202 + 107 = 309	36 + 144 = 180
	13–280	226 + 119 = 345	24 + 24 = 48
Loss through 15 µm sieve during first sieving and second sieving, and loss in coarse material	13–287	734 + 400 + 13 = 1,147	82 + 90 + 6 = 178
	13–288	297 + 1,206 + 13 = 1,516	108 + 220 + 6 = 334
	13–289	418 + 408 + 16 = 842	70 + 156 + 7 = 233
Loss during decanting	13–290	1 + 5 = 6	4 + 6 = 10
	13–291	1 + 8 = 9	14 + 3 = 17
	13–292	1 + 2 = 3	8 + 0 = 8
Average total loss		1,523	364

approximately four times more *Lycopodium* spores are lost during processing than cysts (Table 4).

4. Conclusions and recommendations

- Nine reference samples had estimates of total cyst concentrations ranging from 9106 to 11,815 cysts/g, with an average of 10,691 cysts/g and a standard deviation of 753 cysts/g. Given the range of concentrations, it suggests a random error of approximately 10–15% in absolute cyst abundances.
- Through a thorough investigation of the sieves, stir sticks, coarse fraction, and decanted material used in these reference samples, we document an average loss of 1523 *Lycopodium* spores or 8.2% of the total number of *Lycopodium* spores added per sample with a standard deviation of 323, as well as 364 dinoflagellate cysts or 3.4% of the calculated cyst concentration. Given the loss of *Lycopodium* spores and cysts, our calculated cyst concentrations of the reference samples are overestimated by only ~4.8%. The largest number of cysts and *Lycopodium* spores lost during processing occurs during the sieving process, where they become stuck to the Nylon Nitex sieves and get lost through the 15 µm sieve. Few cysts were recorded in the coarse fraction (>120 µm), stuck to stir sticks, or lost during decanting. Some cysts and *Lycopodium* spores may become entrapped within the mesh itself, but this requires future examination.
- High (>5) cyst/*Lycopodium* spore ratios result in higher standard deviations. If the cyst/*Lycopodium* spore ratio is above 5, we recommend reprocessing the sample adjusting the weight and number of tablets accordingly, or continuing the microscopic analysis until a greater number of *Lycopodium* spores have been counted. Ideally the ratio is between 0.5 and 2.0 to facilitate counting.
- No difference in total cyst concentrations is reported here between 10 µm Nylon Nitex sieves and 15 µm Nylon Nitex sieves.
- Up to two minutes of sonication using a Branson sonic bath does not affect the reproducibility. Longer than 5 min should be avoided.
- Settling for 7–24 h, centrifuging at 3000–3500 rpm, and 6–10 min were all reproducible. We recommend centrifuging at 6 min, as this is most time efficient, and at 3000 rpm to minimize wear on the centrifuge caused by higher speeds.
- Using sediment that has been powdered (ground up) prior to processing should be avoided. It not only results in lower cyst concentrations but also preferentially destroys cysts produced by heterotrophic taxa, thus altering the composition of the original assemblage.
- Adding *Lycopodium* tablets at the end of preparation results in lower concentrations. The “true” cyst concentration is underestimated when *Lycopodium* tablets are added at the end, as cysts are also lost during processing. However, when *Lycopodium* tablets are added at the beginning, cyst concentrations are overestimated as we document four times more *Lycopodium* spores being lost than dinoflagellate cysts. Despite the overestimation that occurs by adding the tablet at the beginning, we recommend using this method as both cysts and *Lycopodium* spores can be assumed to be lost more or less proportionally if some of the sample is accidentally spilled during processing.
- A standard method is proposed (Fig. 4). We recommend using squirt bottles during sieving to expedite the process. If using Nylon Nitex sieves, we also suggest using a magnetic stirrer to prevent residue from settling on the fine sieve.
- Future studies should compare this standardized *Lycopodium* method to alternative methods such as the volumetric method (Dale et al., 2002), the use of microspheres (Ogden, 1986) and non-acid techniques (Riding and Kyffin-Hughes, 2011), to further understand discrepancies between methods.

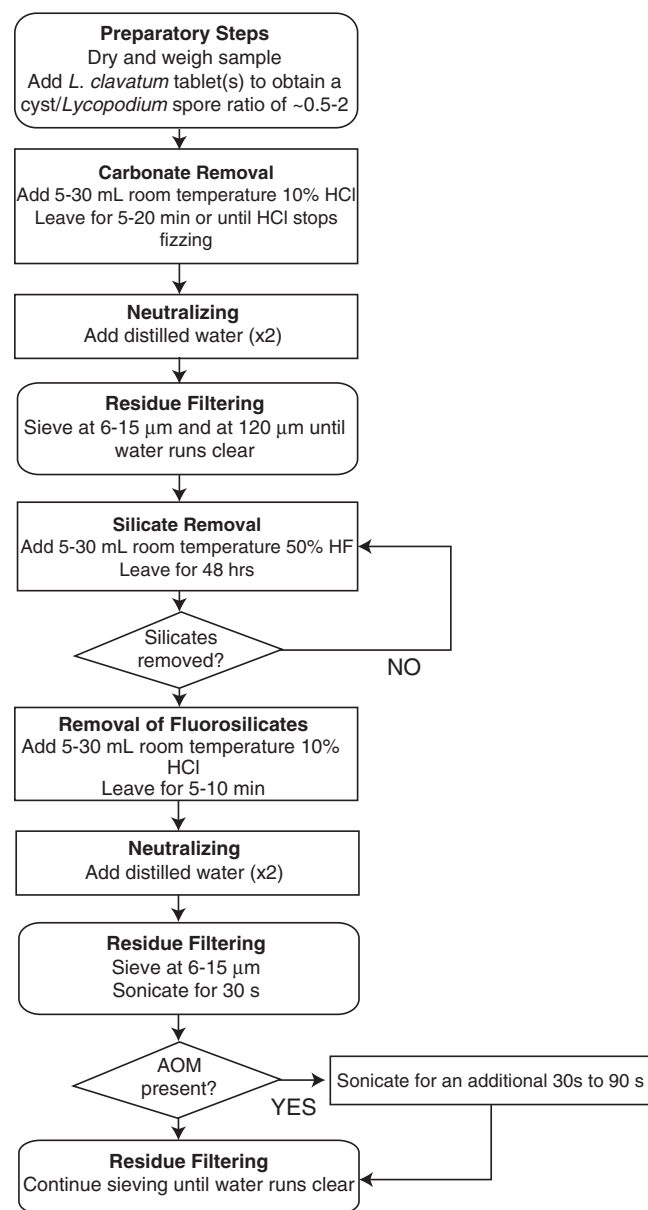


Fig. 4. Flowchart of the proposed standard methodology. AOM stands for amorphous organic matter.

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